

Digital Twin of a 3-DOF Industrial Manipulator

Overview

This project implements a Digital Twin of a 3-DOF planar robotic manipulator using Python. The system simulates robot motion, computes forward kinematics, visualizes trajectories, and exports simulation data.

The objective is to demonstrate how a virtual robot model can replicate the behavior of a physical robotic arm in real time.

Features

- 3-DOF planar robotic arm simulation
- Forward kinematics computation
- Jacobian matrix calculation
- Joint position visualization
- Workspace analysis
- Real-time trajectory animation
- Joint angle plotting
- JSON state export
- Configurable robot parameters

Project Structure

- `kinematics.py` – Forward kinematics, Jacobian, workspace analysis
- `digital_twin.py` – Simulation engine and state management
- `trajectory.py` – Visualization and animation
- `simulation.json` – Configuration and robot parameters

Installation

```
pip install -r requirements.txt
```

Run Simulation

```
python digital_twin.py
```

Run Visualization

```
python trajectory.py
```

Robot Parameters

Parameter	Value
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Parameter	Value
DOF	3
Link 1	110 mm
Link 2	90 mm
Link 3	60 mm
Maximum Reach	260 mm

Future Improvements

- Inverse Kinematics
- Collision Detection
- Obstacle Avoidance
- Path Planning
- Real Hardware Integration